sankalp@sankalp-LOQ-15IRX9:~$ cd TY/SEM\ 6/VSEC/DOCKER$ mkdir ai_docker_project
sankalp@sankalp-LOQ-15IRX9:~$ cd TY/SEM\ 6/VSEC/DOCKER$ cd ai_docker_project
sankalp@sankalp-LOQ-15IRX9:~/TY/SEM 6/VSEC/DOCKER/ai_docker_project$ touch app.py requirements.txt Dockerfile

1.) **USE code into app.py**

```
from flask import Flask, request, jsonify

app = Flask(__name__)

# Dummy AI model (spam detection logic example)
def predict(text):
    if "free" in text.lower():
        return "Spam"
    return "Not Spam"

@app.route('/')
def home():
    return "AI Model is Running 🚀 "

@app.route('/predict', methods=['POST'])
def predict_api():
    data = request.json
    text = data.get("text", "")
    result = predict(text)
    return jsonify({"prediction": result})

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

2) code for requirement:

```
flask
```

3) Code for Docker file(no extension)

```
# Use official Python image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Copy files into container
COPY . /app

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Expose port
EXPOSE 5000

# Run the application
CMD ["python", "app.py"]
```

```
sankalp@sankalp-LOQ-15IRX9:~/TY/SEM 6/VSEC/DOCKER/ai_docker_project$ ls
app.py  Dockerfile  requirements.txt

sankalp@sankalp-LOQ-15IRX9:~/TY/SEM 6/VSEC/DOCKER/ai_docker_project$ code .

sankalp@sankalp-LOQ-15IRX9:~/TY/SEM 6/VSEC/DOCKER/ai_docker_project$ docker build -t
ai_docker_project .
[+] Building 157.4s (9/9) FINISHED                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                    0.0s
 => => transferring dockerfile: 316B                                    0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim       31.7s
 => [internal] load .dockerignore                                       0.0s
 => => transferring context: 2B                                         0.0s
 => [1/4] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd3  116.5s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338  0.0s
 => => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c 251B / 251B  0.9s
 => => sha256:fc74430849022d13b0d44b8969a953f842f59c6e 13.88MB / 13.88MB  17.7s
 => => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a 1.29MB / 1.29MB  8.1s
 => => sha256:38513bd7256313495cdd83b3b0915a633cfa475 29.78MB / 29.78MB  115.6s
 => => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072  0.5s
 => => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa0  0.1s
 => => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c  0.3s
 => => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2  0.0s
 => [internal] load build context                                       0.1s
 => => transferring context: 931B                                       0.0s
 => [2/4] WORKDIR /app                                                  0.1s
 => [3/4] COPY . /app                                                   0.0s
 => [4/4] RUN pip install --no-cache-dir -r requirements.txt              8.1s
 => exporting to image                                                  0.8s
 => => exporting layers                                                 0.5s
 => => exporting manifest sha256:bf0017a0de4e1be597b26f0d2c9a6acabdceefee  0.0s
 => => exporting config sha256:6371fff402fad340707a28a8e0dd797dc17df0034d  0.0s
 => => exporting attestation manifest sha256:5c1dbf550c6ec48f033977b11e3b  0.0s
 => => exporting manifest list sha256:cefa4c555b070f633b83305f8ada98d99a7  0.0s
 => => naming to docker.io/library/ai_docker_project:latest              0.0s
 => => unpacking to docker.io/library/ai_docker_project:latest           0.2s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-
linux/dwbhgmg1wrn1nb3h5gyopuins

sankalp@sankalp-LOQ-15IRX9:~/TY/SEM 6/VSEC/DOCKER/ai_docker_project$ docker run -d -
p 5000:5000 ai_docker_project
8dbeee965d1c17a9adaece8fee013424d70de00697e23dae9397ad4b65b7f05a

sankalp@sankalp-LOQ-15IRX9:~/TY/SEM 6/VSEC/DOCKER/ai_docker_project$ docker ps
CONTAINER ID   IMAGE              COMMAND          CREATED          STATUS          PORTS
NAMES
8dbeee965d1c   ai_docker_project   "python app.py"   42 seconds ago   Up 42 seconds
0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   nervous_nightingale
```

```
sankalp@sankalp-LOQ-15IRX9:~/TY/SEM 6/VSEC/DOCKER/ai_docker_project$ docker stop
8dbeee965d1c
8dbeee965d1c

sankalp@sankalp-LOQ-15IRX9:~/TY/SEM 6/VSEC/DOCKER/ai_docker_project$ docker rm
8dbeee965d1c
```